

UNSUPERVISED ADAPTATION OF AI DOA ESTIMATORS VIA DOWNSTREAM TRACKING

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ABSTRACT

Accurate direction of arrival (DoA) estimation plays a central role in a broad range of applications. While artificial intelligence (AI) has recently emerged as a powerful tool for DoA estimation in challenging scenarios, these AI methods are sensitive to distribution shifts caused by calibration drifts, hardware variations, or changing propagation conditions, and adaptation requires labeled data that may be impractical to obtain during deployment. In this work, we introduce a framework for unsupervised adaptation of AI-based DoA estimators by adopting a holistic system-level perspective. Our key insight is that localization algorithms are rarely applied in isolation, but are instead followed by downstream tracking mechanisms such as Kalman filtering. We exploit the statistical information encoded in the innovations of such tracking algorithms to construct unsupervised performance measures that guide online adaptation of the DoA estimator. We develop a dedicated learning algorithm for continual unsupervised adaptation, and demonstrate through numerical studies that our approach enables reliable and flexible AI-aided DoA estimation in time-varying environments.

Index Terms— Unsupervised learning, DoA estimation.

1. INTRODUCTION

Accurate direction of arrival (DoA) estimation is a fundamental building block in a wide range of modern technologies, including wireless communications, radar, and navigation. Traditionally, this task has been carried out using classic model-based algorithms, such as MUSIC [1] and related subspace methods [2], which exploit the statistical relationship between transmitted sources and the impinging signals. These methods, however, typically rely on simplified assumptions, including fully-calibrated arrays, non-coherent sources, and idealized noise statistics.

The emergence of artificial intelligence (AI) has spurred significant research into data-driven algorithms for DoA estimation. Deep learning models have been shown to enable accurate localization in challenging scenarios where classical methods struggle, such as in the presence of calibration errors, limited snapshots, and low signal-to-noise ratio (SNR) [3]. Existing methods can be broadly categorized into two families. The first employs deep

neural networks (DNNs) such as multilayer perceptrons [4–7], convolutional neural networks (CNNs) [8–11], and attention models [12, 13] to estimate the DoAs from array measurements or related features. The latter follows model-based deep learning [14], integrating DNNs into classical estimators rather than replacing them, thereby combining the interpretability of array processing with the flexibility of learning. Examples include DNNs that augment MUSIC-type methods by providing enhanced spectra [15, 16], or DNNs that generate covariance surrogates for subspace methods to operate under violations of standard assumptions, as in SubspaceNet [17] and related algorithms [18–21].

Despite their success, AI-aided DoA estimators face a critical limitation: they typically assume that the deployment data distribution closely matches the one used during training. In practice, this assumption rarely holds, as the underlying statistical models vary over time and across devices. In principle, such distribution shifts could be handled through online and continual learning of the AI models [22]. However, existing methods are trained in a supervised manner, requiring access to labeled pairs of received signals and their corresponding DoAs. While such ground-truth data can be obtained offline via simulations or measurements, it is unlikely to be provided in real time for online adaptation. Attempts to relax the reliance on labels have led to limited unsupervised formulations. The work [23] proposed to reconstruct the covariance from estimated DoAs, which is valid under simplified statistical assumptions, while [24] translated MUSIC spectra into surrogate unsupervised metrics used for model-based calibration rather than DNN training.

In this work, we propose a framework for unsupervised adaptation of AI-aided DoA estimation methods by adopting a *holistic system-level viewpoint*. While providing an unsupervised assessment of DoA recovery in isolation is inherently challenging, we recognize that such algorithms are rarely deployed standalone. Instead, they are typically integrated within larger systems that incorporate downstream tracking mechanisms, such as Kalman filtering. We exploit this observation by leveraging the statistical information embedded in the innovation features of downstream tracking to guide online, unsupervised adaptation of DNN-based DoA estimators. Building on this principle, we design a dedicated learning algorithm that continuously monitors these innovation features to adjust the DoA estimator to variations in the underlying statistical model. Our numerical study demonstrates that the proposed approach enables reliable and flexible AI-aided DoA estimation under time-varying calibration drifts.

The rest of this paper is organized as follows: Section 2 presents the system model. Our method is derived in Section 3 and evaluated in Section 4. Section 5 provides concluding remarks.

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2. SYSTEM MODEL AND PROBLEM FORMULATION

2.1. System Model

Signal Model: The considered system utilizes a uniform linear array (ULA) comprising N antenna elements to continuously recover the DoAs of M sources. We assume a block-wise operation, where in each block of index i , the system collects T multidimensional snapshots $\mathbf{x}_i(t) \in \mathbb{C}^{N \times 1}$, for $t = 1, 2, \dots, T$, corresponding to M narrowband far-field sources $\mathbf{s}_i(t) \in \mathbb{C}^{M \times 1}$ with DoAs $\boldsymbol{\theta}_i = [\theta_{i,1}, \dots, \theta_{i,M}] \in [-\frac{\pi}{2}, \frac{\pi}{2}]^M$. DoA estimators aim to recover $\boldsymbol{\theta}_i$ from $\mathbf{X}_i = [\mathbf{x}_i(1), \dots, \mathbf{x}_i(T)] \in \mathbb{C}^{N \times T}$.

Under standard far-field propagation, the received $\mathbf{X}_i$ obeys

$$\mathbf{X}_i = \mathbf{A}_i(\boldsymbol{\theta}_i)[\mathbf{s}_i(1), \dots, \mathbf{s}_i(T)] + \mathbf{V}_i, \quad (1)$$

where $\mathbf{V}_i \in \mathbb{C}^{N \times T}$ represents additive noise with distribution function $P_{\mathbf{V}_i}$, while $\mathbf{A}_i(\boldsymbol{\theta}) = [\mathbf{a}_i(\theta_1), \dots, \mathbf{a}_i(\theta_M)] \in \mathbb{C}^{N \times M}$, is the steering matrix. For ideal half-wavelength spaced ULAs, it holds that $\mathbf{a}_i(\theta) = [1, e^{-j\pi \sin(\theta)}, \dots, e^{-j\pi(N-1) \sin(\theta)}]^\top$.

Temporal Variations: The DoAs are assumed to remain static within each block, but vary between blocks due to, e.g., source movement. These variations are modeled as a *state evolution model*, i.e., there exists a function $f(\cdot)$ such that

$$\boldsymbol{\theta}_{i+1} = f(\boldsymbol{\theta}_i) + \mathbf{w}_i, \quad (2)$$

where $\mathbf{w}_i$ is an i.i.d. driving noise with covariance $\mathbf{Q}$.

AI-Aided Localization: We focus on DoA estimation carried out using an AI-aided algorithm, either fully DNN-based, e.g., [8], or a DNN-augmented model-based algorithm, e.g., [17]. Such AI-aided DoA estimators map $\mathbf{X}_i$ into an estimate of the DoAs $\hat{\boldsymbol{\theta}}_i = g_\psi(\mathbf{X}_i)$, where ψ is the trainable DNN parameters.

2.2. Problem Formulation

Our goal is to design an algorithm that enables AI-aided DoA estimators to continuously provide accurate estimates of $\boldsymbol{\theta}_i$ over multiple blocks. Specifically, over a horizon of I blocks, performance is measured in the mean-squared periodic error (MSPE) [25]. We aim to handle the following challenges:

- C1 Nonstationarity:* The statistical model (1) (e.g., the statistics of the noise $P_{\mathbf{V}_i}$ can change, and the array response $\mathbf{a}_i(\theta)$) can change between blocks, for instance, due to calibration drifts. Unlike the variations in the DoAs, which are modeled via (2), we do not assume any prior knowledge of the nature of the variations in statistical model.
- C2 Unlabeled Data:* The DoA estimator $g_\psi(\cdot)$ is pre-trained offline using labeled data (array measurements and DoAs) from the initial block ($i = 1$). However, once deployed, it has no access to ground-truth DoAs at any block $i > 1$.

3. UNSUPERVISED ONLINE LEARNING

Adapting AI-aided DoA estimators in deployment faces a tension between challenges *C1* and *C2*. Handling *C1* with undisclosed

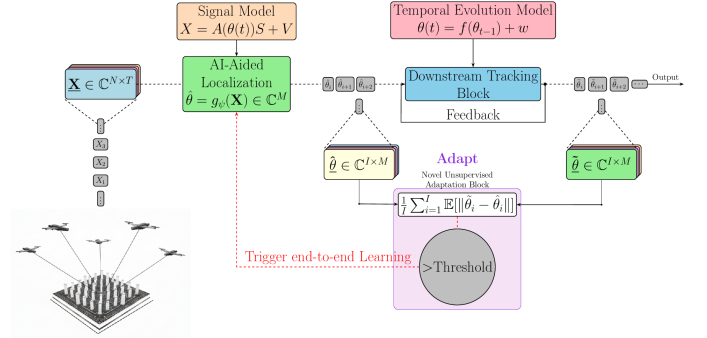


Fig. 1: System architecture illustrating the signal model, AI-aided DoA estimation, downstream tracking, and the adaptation mechanism triggered by innovation statistics.

changes in the *current* signal statistics requires ground-truth data to refine trainable parameters ψ . Yet this need clashes with *C2*, which rules out ground-truth DoAs during operation.

Our approach resolves this tension by leveraging the temporal evolution modeled in (2). We thus focus on the expected operation in which the DoA estimator is not a standalone component but is followed by a tracking algorithm that exploits the Markovian state evolution, which is monitored as a measure for online adaptation. We next describe the overall system (Subsection 3.1) and the resulting online learning algorithm (Subsection 3.2) followed by a discussion (Subsection 3.3).

3.1. System Overview

The overall pipeline (Fig. 1) consists of four modules:

- 1. AI-aided localization:** For each block i , the AI-aided estimator maps the array measurements to a DoA estimate, $\hat{\boldsymbol{\theta}}_i = g_\psi(\mathbf{X}_i)$.
- 2. Downstream tracking:** A Bayesian tracker (e.g., Kalman-type) exploits the temporal model in (2) to update the instantaneous estimate $\hat{\boldsymbol{\theta}}_i$ into the system prediction $\hat{\boldsymbol{\theta}}_i$ and to filter consecutive estimates across blocks.
- 3. Innovation monitoring:** Our main measure for adaptation is based on monitoring the tracking innovation term:

$$\mathbf{y}_i \triangleq \hat{\boldsymbol{\theta}}_i - g_\psi(\mathbf{X}_i). \quad (3)$$

The innovation $\mathbf{y}_i$ in (3) is computed and summarized over sliding windows to form statistics used for drift detection.

- 4. Adaptation:** When the monitored statistics indicate mismatch, an unsupervised loss drives gradient-based updates of ψ over a short adaptation horizon, after which online inference resumes with the updated model (see Subsection 3.2).

Monitoring $\{\mathbf{y}_i\}$ over multiple blocks yields an unsupervised signal that: *i*) detects distribution shifts caused by calibration drifts or environmental changes (*C1*); and *ii*) guides adaptation without labels (*C2*). In this way, the AI-aided DoA estimator is updated based solely on consistency between its outputs and the downstream tracking, obviating the need for ground-truth DoAs.

3.2. Unsupervised Learning Algorithm

We continuously assess the agreement between the data-driven AI predictions and the temporally consistent tracking using

statistics of the innovation sequence $\{\mathbf{y}_i\}$. To that aim, over a sliding window of I blocks, we compute

$$\sigma_y^2(i) = \frac{1}{I} \sum_{j=i-I+1}^i \|\mathbf{y}_j\|^2. \quad (4)$$

The empirical variance $\sigma_y^2(i)$ captures short-term consistency between the AI estimator and the tracker. Adaptation is triggered when the innovation statistics depart from nominal values, e.g., if $\sigma_y^2(i) > \tau_\sigma$ with thresholds τ_σ . When $\{\mathbf{y}_i\}$ is modeled as zero-mean i.i.d. Gaussian, this implements a form of Chi-Squared test for which the threshold can be computed analytically. However, in our numerical examples we tune τ_σ manually to avoid imposing a distribution on (3).

When it is triggered that adaptation is required, we update the parameters of the AI-aided estimator ψ using an innovation-based loss we term *mean-squared innovation error (MSIE)*. This loss, defined over a window $\mathcal{W}_i$ of I blocks indexed $i - I + 1, \dots, i$, is given by

$$\mathcal{L}_{\mathcal{W}_i}(\psi) = \frac{1}{I} \sum_{j=i-I+1}^i \|\tilde{\theta}_j - g_\psi(\mathbf{X}_j)\|^2. \quad (5)$$

Parameters are updated by gradient descent (GD) on $\mathcal{L}_{\mathcal{W}_i}(\psi)$. The complete online procedure is summarized in Algorithm 1. While the formulation in Algorithm 1 focuses on a single time block i for simplicity, when adaptation is triggered (i.e., $\sigma_y^2(i) > \tau_\sigma$), we repeat the GD steps also over the subsequent W windows, after which online operation continues with the updated ψ .

Algorithm 1: Unsupervised Adaptation at block i

Init : Window size I ; threshold τ_σ ;
step-size η ; GD steps S ;

Input : Observations $\{\mathbf{X}_i\}$, current DNN weights ψ

- 1 Compute $\hat{\theta}_i = g_\psi(\mathbf{X}_i)$;
 - 2 Obtain tracking prediction $\tilde{\theta}_i$ from downstream filter;
 - 3 Compute tracking innovation $\mathbf{y}_i$ via (3);
 - 4 Update $\sigma_y^2(i)$ via (4);
 - 5 **if** $\sigma_y^2(i) > \tau_\sigma$ **then**
 - 6 Form the window $\mathcal{W}_i = \{(\mathbf{X}_j, \tilde{\theta}_j)\}_{j=i-I+1}^i$;
 - 7 **for** $s = 1$ **to** S **do**
 - 8 Compute $\mathcal{L}_{\mathcal{W}_i}(\psi)$ using (5);
 - 9 Update $\psi \leftarrow \psi - \eta \nabla_\psi \mathcal{L}_{\mathcal{W}_i}(\psi)$;
 - 10 **return** ψ
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3.3. Discussion

The proposed method is particularly suitable for tackling the problem as formulated in Section 2. By adopting a system-level viewpoint and integrating downstream tracking, we resolve the inherent tension between nonstationarity (C1) and the lack of online labeled data (C2). Our design combines unsupervised

adaptation with a mechanism that triggers adaptation as a form of concept drift detector [26, 27]. Our detector does not monitor changes in the impinging signals, but instead focuses on variations in the outputs of the AI-aided estimator. In this way, it leverages the ability observed for AI-aided DoA estimators to generalize to some forms of distribution changes [3], ensuring that adaptation is only activated when needed, thus avoiding unnecessary updates in response to distributional changes that do not degrade performance substantially.

Algorithm 1 facilitates adaptive and autonomous operation at the cost of excessive complexity compared to using static pretrained systems. In particular, as its test statistic is based on a simple empirical squared deviation mean (5), its complexity is dominated by training procedure conducted once adaptation is triggered. The latter involves taking $W \cdot I \cdot S$ gradients, i.e., back-propagation steps through $g_\psi(\cdot)$. Its excessive burden can thus be balanced by setting the hyperparameters W, I, S , as well as by making adaptation less frequent by increasing τ_σ . In our experimental study we tune these hyperparameters manually. However, a potential follow-up direction is to employ downstream features not only for constructing unsupervised training losses, but also for guiding the adaptation of hyperparameters such as step sizes and window lengths in an online fashion. Another promising extension is to consider alternative test statistics for triggering adaptation. These extensions are left for future research.

4. NUMERICAL STUDY

4.1. Experimental Setup

In this section we numerically evaluate Algorithm 1 for online adaptation of AI-aided DoA estimators¹. We consider a ULA comprising $N = 9$ antenna elements operating at 5 GHz with DoAs uniformly generated from $[-\frac{\pi}{2}, \frac{\pi}{2}]$. We evaluate two snapshot regimes: $T = 200$ and $T = 20$, representing high and limited observation scenarios. The impinging signals are narrowband and non-coherent originating from $M = 3$ sources.

Three sources move over 100 blocks following with an evolution model (2) given by $\theta_{i+1} = \theta_i + [3, -3, 2] \odot \sin(\omega_0 \cdot t) + \mathbf{w}_i$, where $\omega_0 = [-0.15, 0.25, 0.15]$ and $\mathbf{w}_i$ is i.i.d. Gaussian noise with standard deviation 0.03. Distribution shifts are introduced at a randomly selected block to simulate calibration drifts encountered in deployment. These shifts model two types of array imperfections: (i) **Distance miscalibration**, representing deviations from nominal half-wavelength spacing, modeled as uniform perturbations $\delta_n \sim \mathcal{U}(-\eta, \eta)$ with $\eta \in \{0.3, 0.9, 1.2\}$ that affect the phase response; and (ii) **Geometric noise**, representing additional steering vector perturbations, modeled as additive complex Gaussian noise $\epsilon_{n,\theta}$ with variance 0.9. The combined effect produces a perturbed steering matrix:

$$[\mathbf{a}(\theta)]_n = e^{-j2\pi \frac{(d_n + \delta_n) \sin(\theta)}{\lambda}} + \epsilon_{n,\theta}, \quad (6)$$

¹The source code and all hyperparameters are available online at <https://github.com/UliKonstantin/SubspaceNet>

SNR [dB]	Pre-trained		Online supervised		Algorithm 1	
	$T = 20$	$T = 200$	$T = 20$	$T = 200$	$T = 20$	$T = 200$
0	11.3°	11.15°	5.47°	5.38°	7.90°	7.51°
5	12.37°	12.31°	5.65°	5.57°	7.95°	8.80°
10	17.44°	16.06°	4.54°	4.37°	6.51°	6.47°

Table 1: Time-averaged RMSPE for different SNRs, after distribution change due to calibration shift

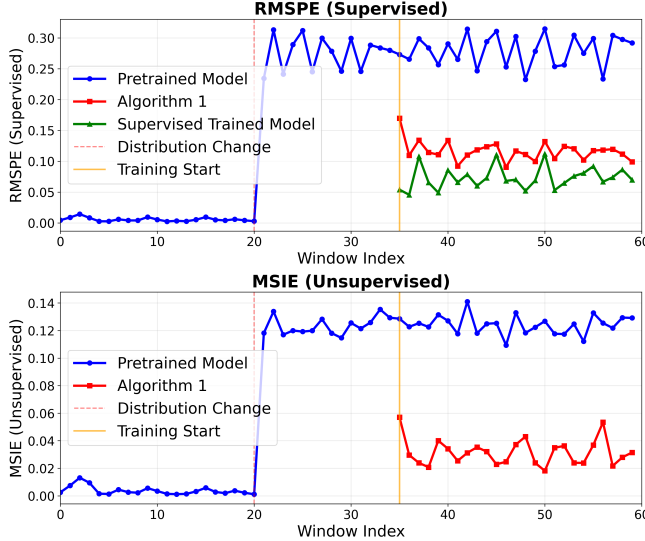


Fig. 2: Performance in RMSPE (upper plot) and MSIE (lower plot) versus block index, $T = 200$ snapshots per block

where the distance drift δ_n disturbs the steering matrix through phase shifts, and the geometric noise $\epsilon_{n,\theta}$ is additive to the steering matrix itself. As we model shifts in the array response, we keep the noise $\mathbf{V}_i$ in (1) stationary, and set its distribution to be Gaussian.

As our AI-aided DoA estimator, we use the SubspaceNet algorithm of [17], whose DNN ψ is based on a convolutional autoencoder. SubspaceNet is pre-trained on 4096 labeled samples from an ideal ULA. The downstream tracking algorithm is a Kalman filter with plugged in state evolution noise variance and observation variance of 0.03. In implementing Algorithm 1, we consider a sliding window of $I = 5$ blocks, adaptation period of 10 windows, and adapt using $S = 5$ GD steps when triggered. We also evaluate supervised online learning which requires one to provide ground-truth DoAs online once the distribution shifts.

4.2. Numerical Results

We first evaluate the time-averaged rmspe over all 300 blocks of length $T = 200$ and $T = 20$ and averaged over 100 Monte Carlo trials. In Table 1 we compare the results achieved by tracking the DoA based on the pre-trained static SubspaceNet to those achieved with our unsupervised Algorithm 1 as well as with (impractical) online supervised learning. The algorithms are compared different SNR values, defined here as the inverse variance of $\mathbf{V}_i$. Table 1 demonstrates the online learning model consistently outperforms the pre-trained baseline across all SNR conditions. The pre-trained model struggles coping with distribution shifts, particularly trained at data with higher SNRs.

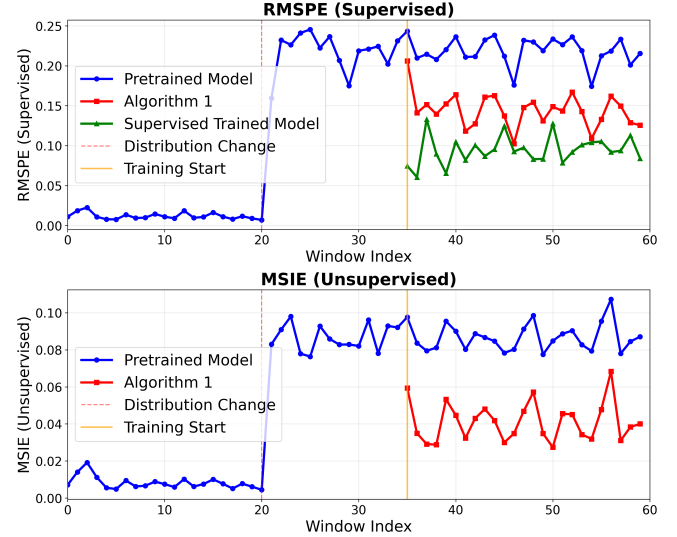


Fig. 3: Performance in RMSPE (upper plot) and MSIE (lower plot) versus block index, $T = 20$ snapshots per block

Accordingly, Algorithm 1 demonstrates substantial gains ranging from 3.5° to 14.46° and approaches the performance that is achieved with supervised online training.

We proceed to demonstrating the ability of Algorithm 1 in rapidly mitigating the effects of distribution shifts by monitoring performance per block. Specifically, we report both the actual error of the overall system (i.e., the root MSPE (RMSPE)) and the unsupervised loss utilized by Algorithm 1, namely, the MSIE (5). We monitor these figures for a scenario with relatively long blocks ($T = 200$) as well as limited ($T = 20$) snapshot conditions at varying SNR of [0, 5, 10] dB, with the detailed evolution curves reported in Figs. 2 and 3, respectively. Observing Figs. 2-3, we note clear adaptation phases where the online unsupervised model progressively improves performance, and is only within a minor gap of the performance achieved with supervised online learning. The unsupervised MSIE metric effectively tracks supervised RMSPE performance, validating that innovation statistics serve as reliable proxies for estimation quality. Notably, the online adaptation presents high tolerance to resource-constrained scenarios with limited observations. These results validate the effectiveness of our unsupervised adaptation framework, demonstrating maintained accurate DoA estimation under challenging conditions including low SNR, limited observations, and model miscalibrations, without requiring labeled data during deployment.

5. CONCLUSIONS

We proposed an unsupervised adaptation framework for AI-aided DoA estimators that leverages downstream tracking innovations to enable online learning without labeled data. Our approach successfully handles distribution shifts caused by calibration drifts and environmental changes while maintaining accurate estimation performance. Numerical results demonstrate substantial improvements, validating the effectiveness of our method for robust DoA estimation in time-varying environments.

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